

John Seungjin Oh

U.S. citizen searching for opportunities to satisfy my curiosity and passion for problem-solving.

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Education

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Computer Science; Additional Major in Philosophy

AUG 2023 - MAY 2027

- **Selected Coursework:** Parallel Data Structures and Algorithms, Machine Learning, Causal Machine Learning, Interactive Extended Reality, Topological Philosophy of Science, Computer Systems

Technical Skills

Awards: AIME Qualification, USACO Gold, Scholastic Writing Awards National Gold Medal

Languages/Frameworks/Technologies: JavaScript/TypeScript, Python, Flask, Go, C, Java, BigQuery, AWS, GCP, GraphQL, Node.js, React.js/Next.js, C#, SQL, Docker, Linux, Git, Bash, LangChain/LangGraph, Langfuse, SML

Other: Fluent Korean, Intermediate Spanish, Growthbook, Prompt Engineering, Causal inference/discovery, Claude Code, Cursor, VM VirtualBox, Wireshark, Unity, Latex

Experience

Khan Academy

Mountain View, California

Backend Software Engineer Intern

MAY 2025 - AUG 2025

- **Led** the "Memory Evaluation" project, which was Khan Academy's first attempt to build offline cross-thread LLM evals for Khanmigo, our core AI tutoring assistant. Designed novel scoring functions in Golang to compare LLM behaviors and prompts, evaluated various "memory mechanisms", and researched strategies to enable scalable human-labeling of cross-thread data.
- **Successfully handled** a major end-to-end migration from our internal OpenAI moderation client to an open source Go OpenAI moderation client; unlocked new moderation capabilities and enhanced LLM observability tooling via LangFuse
- **Spearheaded** the intern hackathon project, which aimed to support user-generated and AI-generated flashcards embedded in the Khan Academy website. **Designed** the GraphQL backend architecture, implemented LaTeX rendering for flashcards, and successfully presented work to top stakeholders. Got a thumbs up from Sal Khan!

Alstom

Pittsburgh, Pennsylvania

Software Intern

MAY 2024 - FEB 2025

- **Standardized** configuration variable management for automatic train control software by developing an internal tool using Next.js and Python, streamlining version control among teams worldwide and massively reducing the time spent on tedious configurations.
- **Built** an end-to-end solution for compactly defining and using proprietary network protocols. Created a custom protocol definition schema which could be used to generate Lua dissectors for arbitrary protocols. Developed a custom frontend interface for Wireshark that can dynamically dissect specified protocols and assist with packet data analysis.
- **Prototyped** an AI-powered tool that used LLM agents to allow users to perform text-to-SQL and analyze complex log data using natural language. Researched the use of local LLMs to address concerns in privacy, cost, and accessibility.
- **Implemented** computer vision algorithms using OpenCV to automate GUI testing with the Robot Framework

Whee School Inc.

Philadelphia, Pennsylvania

Software Engineer

DEC 2023 - MAY 2024

- **Conducted** a partial migration of the landing website and LMS (learning management system) of Whee School from HTML/CSS/Javascript to modern Next.js/Tailwind/TypeScript with better UI/UX design and code maintainability
- **Managed** relevant AWS technologies, including CodePipeline, RDS, IAM, and ECS
- **Developed** a new service that incorporated OpenAI API and advanced prompting techniques to help teachers analyze their students' writing

Open Source

PrimeIntellect (Prime Environments)

OCT 2025 - PRESENT

- **Contributing** a new Document Search RL environment for agentic search with patent data.
- **Created** a (12k+) AR/VR/MR-focused patents corpus by post-processing and filtering a pre-existing corpus.
- **Created** a high-quality synthetic Q&A dataset for patents.